

PHYSICOCHEMICAL PROPERTIES OF GUM ARABIC

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ABSTRACT

Gum Arabic is harvested commercially from wild trees throughout the Sahel from Senegal and Sudan to Somalia and in the northern part of Nigeria. Clumps of gum Arabic were collected and dirt particles were removed, the samples were dried and grounded to fine powder. The Physicochemical properties of gum Arabic was determined using standard methods and the values obtained showed; Moisture content $6.9\% \pm 5$, Soluble content 90.6%, Viscosity $5.45 \pm 3 \text{Ns/m}^2$, Ash content $3.2 \pm 4\%$, Zinc 3mg/kg, Iron 41mg/kg ± 5 , Manganese 48.2mg/kg ± 5 and Copper 33.3mg/kg ± 2 . The gum also contains carbohydrate 0.3ppm ± 2 , protein 0.75ppm ± 2 , starch 0.0076 ± 10 and nitrogen 0.12ppm ± 5 . It was observed that the gum does not contain cadmium and nickel.

Keywords: Gum Arabic, physicochemical, hydrocolloid

INTRODUCTION

Acacia Senegal, popularly known as gum Arabic is a leguminous tree crop which belongs to the family Mimosaceae. This family contains over 300 known species including Acacia Senegal and Acacia Seyal and are the most commercially exploited species. Both species are considered as the important economic Acacia plant in the Sudan-Saharan ecological zones covering not less than eleven states in the northern part of Nigeria. The species produces the best grade of gum in commercial quantity in Nigeria and it is an important source of foreign exchange earnings (Aghughu, 1998).

Acacia gum is the dried exudates obtained by tapping the stems and branches of acacia Senegal or other related species. The gum has various uses in the food, textiles, prints and

pharmaceutical industries in the world. The trees are planted in shelterbelt to check wind and soil erosion, hence it is used to fight against desert encroachment (Aghughu, et al., 2005). The plant adds nitrogen to the soil through its nitrogen fixing ability and leaf litter fall thereby improving soil fertility (Cossalter, 1991). Its pods, seeds and leaves are excellent browse and folder for livestock (Scholte, 1992).

Gum Arabic is harvested commercially from wild trees throughout the Sahel from Senegal and Sudan to Somalia, although it has been historically cultivated in Arabia and West Asia. Gum Arabic is a complex mixture of polysaccharides and glycoproteins that is used primarily in the food industry as a stabilizer. It is edible and is a key ingredient in traditional lithography and used in printing, paint production, glue, cosmetics and various industrial applications, including viscosity control in inks and in textile industries, although less expensive materials compete with it for many of these roles. (Smolinske, 1992).

Gum Arabic is unique among the natural hydrocolloid because of its extremely high solubility in water. Most common gums cannot be dissolved in water at concentration higher than about 5% because of their very high viscosities. Gum Arabic is insoluble in oils and in most organic solvents. Whereas most gums form highly viscous solution at low concentration of about 1-5%, high viscosities are not obtained with gum Arabic until concentration of about 40-50% is obtained.

Technically, gum Arabic is classed in a group of substances called arabinogalactan protein. Eighteen different amino acids have been identified in acacia Senegal, although only four comprises more than 10% of the protein. The gum also comprises essentially a complex polysaccharide, consisting mostly of galactose, arabinose, rhamnose and glucuronic acid (William and Idris, 2000). Gum Arabic is useful as hydrocolloid, emulsifier, texturizer and film-former (Phillip and Ogasawara, 2007).

Substances frequently called gums are hydrocarbons of high molecular mass; others are petroleum products, rubber latex, synthetic polymeric gums, balms and resins. Recently, the term “gum” as technically employed in the industry, refers to plant or microbial polysaccharides and their derivatives that are capable of forming dispersions in cold or hot water, producing viscous mixtures or solutions (De Pinto et al., 1995).

MATERIALS AND METHODS

SAMPLING

The sample was harvested from Acacia Senegal. The stem of the specie was cut and the gum oozed out and dried at room temperature of $30^{\circ}\text{C}\pm 5$ to form clumps. The clumps were collected and particles such as sand, sticks and dirt were removed from the clumps and preserved with formaldehyde to avoid micro organism infestation which may cause biodegradation of the gum. The samples were dried for easy grounding. The dried samples were grounded to fine power and kept in an air-tight container.

Materials used include; Ethanol, tetraoxosulphate (VI) acid, toluene, anthrone reagent, hydrochloric acid, sodium carbonate, alkaline sodium phenate, sodium hypochlorite, sodium potassium tartrate, dry ammonium sulphate and sodium sulphate. They were supplied and used as standard analytical reagents.

METHODS

Metal contents was determined using the Atomic Absorption Spectrophotometer (AAS), Sugar contents and Carbohydrate contents were determined using Anthrone method, Viscosity was determined using Oswald Viscometer, pH was determined using the Jenway 3510 pH meter. Other properties such as Ash content, Moisture content, Insoluble matter, Soluble matter, Nitrogen and Protein were determined using standard methods.

RESULTS AND DISCUSSION

RESULTS

Table I: Results of physio-chemical properties

Introduction Tutorial Questionnaire – Skip Logic

<u>PROPERTIES</u> Single————Choice	<u>VALUES</u>
Questions	
<u>Ash Content</u>	<u>3.23%</u>
<u>Moisture Content</u>	<u>6.9%</u>
<u>Insoluble Matter</u>	<u>9.4%</u>
<u>Soluble Matter</u>	<u>90.6%</u>
<u>Viscosity</u>	<u>5.45Ns/m²</u>
<u>pH</u>	<u>8.16</u>

Table II: Results of Organic Parameters

A1. Do you like colors?—

Yes, No—

[Single-select]

A1A. What color is a ripe tangerine?

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow

[Single-select; split in 2 columns]

A1B. What color is a ripe banana?

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow

[Single-select drop-down menu]

A1C. What color is a ripe strawberry?

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow

[Single-select with horizontal scale]

[Terminate if two or more: A1A is not 4 (orange) / A1B is not 8 (yellow) / A1C is not 6 (red)]

<u>PARAMETERS</u> Multiple————Choice	<u>VALUES</u>
Questions	
<u>Carbohydrate</u>	<u>0.275ppm</u>
<u>Sugar</u>	<u>0.0081ppm</u>
<u>Nitrogen</u>	<u>0.1200ppm</u>

<u>Protein</u>	<u>0.7500ppm</u>
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Table III: Results of Metal Contents Analysis

[Skip A2 if option 2 (no) is selected at A1]

A2. Of these, what colors do you like? **[Select all that apply]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow

[Allow multiple selection]

[Skip A2A if options 3 (green), 6 (red), 8 (yellow) are NOT selected at A2]

A2A. What 3 colors are most common for apples? **[Select exactly 3]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow

[Multi-select, shown in 4 columns]

A2B. What colors are grapes? **[Select all that apply]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow **NEW:** Other (specify)

[Multi-select, with 'other specify' option]

[Terminate if option 3 (green) or 6 (red) are not selected at A2B]

A2C. What colors are in a rainbow? **[Select all that apply]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow **NEW:** None

[Multi-select, with exclusive 'none' option]

[Terminate if option 99 (none) is selected at A2C]

[Skip A2D if option only 1 option selected at A2]

A2D. Are any rooms in your home painted in these colors you like? **[Select all that apply]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow **NEW:** None

[Multi-select, exclusive 'none' option; Show options checked at QA2]

A2E. Are any rooms in your home painted any of these colors? **[Select all that apply]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow **NEW:** None

[Multi-select, exclusive 'none' option; Show options not checked at QA2D]

[Skip A2F if option 99 (none) is selected at A2D AND option 99 (none) is selected at A2E]

A2F. What color decor accents do you prefer? **[Hold ctrl, scroll or click to pick up to 4]**

Black, Blue, Green, Orange, Purple, Red, Violet, Yellow **NEW:** White, Brown, Grey

[Multi-select, with new local options. Only allow up to 4 colors]

<u>METALS</u> Text Entry Questions	<u>VALUES (mg/kg)</u>
<u>Iron</u>	<u>41.0</u>
<u>Manganese</u>	<u>48.2</u>
<u>Copper</u>	<u>31.3</u>
<u>Zinc</u>	<u>3.0</u>
<u>Cadmium</u>	<u>ND</u>

Nickel	ND
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ND means “Not Detected”

DISCUSSION

The results of the physio-chemical properties of gum Arabic (from acacia Senegal) are displayed in Table I. It shows ash content, 3.23%, moisture content, 6.9%, insoluble matter, 9.4% soluble matter, 90.6%, viscosity, 5.45Ns/m² and pH, 8.16. Gum Arabic tends to be more viscous than that of cashew tree exudates. The viscosity of cashew tree exudate is 3.64Ns/m². The reason is that gum Arabic has less moisture contents of 6.9%, compared to that of cashew gum which has a moisture content of 9%. This suggests that acacia tree tends to store much moisture on its body for survival, unlike cashew tree.

From the standard in literature review by Willy Benecke, the ash content in gum Arabic is 4%. This has a close value with the result from this research, with a difference of 0.8% approximately. The standard insoluble matter is 1% which is not too close to the result of insoluble matter, 9.4%. The reason could be that the actual specie of gum Arabic used for the standard was not specified because physiochemical properties vary from specie to specie.

Table II displays the content of the organic nutrient in the sample; gum Arabic. It reveals that carbohydrate 0.275ppm, sugar content 0:00816ppm, Nitrogen 0.1200ppm and protein content is 0.7500ppm. Organic parameters seems to be more in cashew tree exudates which has carbohydrate content as 0.5674ppm, protein 1.0200ppm, nitrogen content 0.1640 and sugar content 0.6280ppm. The reason is traced to the different regional areas of their habitation. Nutrients are easily lost to the environment especially in savanna region where acacia trees inhabits. These values give the gum a perfect option to emulsify.

The results from the metal content analysis is given in Table III, which shows iron content; 40mg/kg, manganese; 48.2mg/kg, copper; 31.3mg/kg and Zinc; 3.0mg/kg. Cadmium and nickel was found to be absent. The absence of these two poisonous metals makes the gum applicable in the food industry.

CONCLUSION

Gum Arabic is unique among the natural hydrocolloids. Gum Arabic's mixture of saccharides and glycoproteins gives it the properties of a glue and binder which is edible by humans. For artists, it is the traditional binder used in water colour paint, in photography for gum printing and as a binder in pyrotechnic compositions. It has been investigated for use in intestinal dialysis. Pharmaceuticals and cosmetics also use the gum as a binder, emulsifying agent and suspending or viscosity increasing agent (Smolinske, et al.,1992). These applications are successfully achieved on the basis of the physicochemical properties which have been established by this research work. The gum is highly viscous and slightly basic and contains little carbohydrate and sugar compared to protein and nitrogen content, suggesting it to be a leguminous plant. Manganese, zinc, iron and copper are the analyzed trace metal content. These trace metals are vital to the proper growth and development of the acacia plant (Svehla, 2007).

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A3. What color is your hair?

[PROVIDE TEXT BOX] Enter natural color

[Terminate if 'green' entered at A3]

A3A. What is the most common color you dye your hair?

[Enter color or select 'I do not dye my hair']

Preferred hair dye color: _____ / I do not dye my hair **[EXCLUSIVE]**

A3B. How many times a day do you wash your hair?

[Enter a number 0 to 5]-

[FORCE NUMBER] _____ time(s) per day

[Ask follow up questions A3C-A3G if 1 or more entered at A3B]

A3C. For classification, what is your zip code?

[ZIP FORMAT] Zip code: _____

A3D. Where can we email our hair care instructions?

[EMAIL FORMAT] **[Enter your email]** _____

A3E. Best number to reach you?

[PHONE FORMAT] Phone number: _____

A3F. What is/was the date of your appointment?

[DATE FORMAT] **[Enter or select date]** _____

A3G. Please explain how you wash your hair in full detail.

[Please type in a response]

Table Rows

B1A. How common or rare is each shape?

Options: 1=Common to 10=Rare

[One per row] Circle, Square, Triangle, Rectangle, Trapezoid

[Terminate if selected option 10 (rare) for each shape at table B1A]

B1B. How many of each shape do you typically encounter per day?

Options: None, 1-5, More than 5

[One per row] Circle, Square, Triangle, Rectangle, Trapezoid

B1C. Do you talk about shapes in any of these places?

Options: At work, At home, In nature, The city

[All per row] Circle, Square, Triangle, Rectangle, Trapezoid

[Include opt-out option] I don't discuss them

[Skip B1D if None for all shapes at B1B]

B1D. Do you see shapes in any of these places?

Options: At work, At home, In nature, The city **NEW:** I don't

[All per row] Circle, Square, Triangle, Rectangle, Trapezoid

B2. Indicate your preference for each pair of statements.

[Select the level that closest fits you]

Option scale: < Fits Me, Equal, Fits Me >

[RANDOMIZE ROWS]

Large shapes / Small shapes

Solid shapes / Shape outlines

3D shapes / 2D shapes

Curved shapes / Angled shapes

Irregular shapes / Semitic shapes

Table Columns

B3A. Which attributes do you associate with each shape?

[Check all that apply for each shape]

Across: Circle, Square, Triangle, Rectangle, Trapezoid

Down: Useful tool, Pleasing lines, External boundaries, None of these

B3B. Select two features that most describe each shape.

[Check two for each shape]

Across: Circle, Square, Triangle, Rectangle, Trapezoid

Down: Shapely, Geometric, Uniform, Precise, Symbolic

B4. Enter your preferences for each shape you use.

[Select yes/no to indicate use]

[ASK COLOR AND SIZE FOR EACH SHAPE WITH YES]

Color: Black, Blue, Green, Orange, Purple, Red, Violet, Yellow

Size: [numeric entry in inches]

Table Features

B5A. Select the properties that you like about each shape.

Options: Angle(s), Point(s), Line(s), Curve(s)

Circle, Square, Triangle, Rectangle, Trapezoid **NEW:** Ellipse, Crescent, Star, Heart

[ONLY ALLOW RELEVANT PROPERTIES FOR THE SHAPES: NO angles/points with circles/ellipses; curves ONLY with circles/ellipses/crescents/hearts]

B5B. Do you know the formula to calculate the area of each shape?

Options: Yes, No, N/A

Circle, Square, Triangle, Rectangle, Trapezoid **NEW:** Ellipse, Crescent, Star, Heart

[INCLUDE OTHER SPECIFY]

B5C. What percent of space would you use for each shape? **[Must total 100%]**

[SHOW TOTALING ROW]

Circle, Square, Triangle, Rectangle, Trapezoid

B5D. What shapes come to mind?

[CAPTURE RESPONSE; ALLOW UP TO 10 SHAPES]